Four minimal relations in a nearly Gorenstein numerical semigroup

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Abstract

The numerical semigroup $S = \langle 6, 7, 8, 9\rangle$ is nearly Gorenstein, is not almost symmetric, and has exactly four minimal defining relations. It therefore gives a negative answer to the assertion $\rho(S) \in \{5, 6\}$ in Problem 23 of version 3 of the survey by Moscariello and Sammartano. The example and its four-relation presentation are already in the literature. We give a self-contained verification of every hypothesis, an elementary proof of generation and minimality of the defining ideal, and an independent exact certificate using factorization graphs. We also distinguish the survey's formulation from the original question of Moscariello and Strazzanti, which expressly includes four relations in type two.

1 The statement and its sources

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. A numerical semigroup is an additive submonoid $S \subseteq \mathbb{N}$ with finite complement. Its embedding dimension is the number of minimal positive generators. Write

$$F = \max(\mathbb{Z} \setminus S), \qquad \mathrm{PF}(S) = \{f \in \mathbb{Z} \setminus S : f + s \in S \text{ for every } s \in S \setminus \{0\}\}.$$

The type is $t(S) = |\mathrm{PF}(S)|$. For minimal generators $g_1, \ldots, g_e$, the semigroup is *nearly Gorenstein* if there are $p_i \in \mathrm{PF}(S)$ such that

$$g_i + p_i - q \in S \qquad (1 \le i \le e,\ q \in \mathrm{PF}(S)). \tag{1}$$

It is *almost symmetric* if every $a \in \mathbb{Z} \setminus S$ satisfies either $F - a \in S$ or $a, F - a \in \mathrm{PF}(S)$.

Let I_S be the kernel of the homomorphism $\mathbb{Q}[x_1, \ldots, x_e] \longrightarrow \mathbb{Q}[t]$ sending x_i to t^{g_i}, and let $\rho(S)$ be the minimum number of binomials generating I_S. Problem 23 in [1, version 3] asks whether

$$e = 4, \quad S \text{ nearly Gorenstein and not almost symmetric} \quad \Longrightarrow \quad \rho(S) \in \{5, 6\}. \tag{2}$$

There is a material difference between this formulation and its cited source. Moscariello and Strazzanti [2, Question 4.3] ask whether type two gives four or five relations, and type three gives six. Their Proposition 3.4 also places semigroups generated by an arithmetic sequence in the nearly Gorenstein class. Gomes, O'Neill, and Torres Davila [3, Examples 8 and 18] give the Apéry set and a four-relation minimal presentation of $\langle 6, 7, 8, 9\rangle$ explicitly. Thus the example below is a known case, and its relation count belongs to the type-two branch of the original question. The purpose of this note is to make its consequence for (2) explicit and fully verifiable.

Theorem 1. *The semigroup $S = \langle 6, 7, 8, 9\rangle$ has embedding dimension four, $F = 11$, and $\mathrm{PF}(S) = \{10, 11\}$. It is nearly Gorenstein and is not almost symmetric. Under $(x, y, z, w) \mapsto (t^6, t^7, t^8, t^9)$, its defining ideal is*

$$I_S = (y^2 - xz,\ yz - xw,\ z^2 - yw,\ w^2 - x^3), \tag{3}$$

and every ideal-generating set has at least four elements. In particular, $\rho(S) = 4$, so (2) is false.

2 The semigroup hypotheses

The numbers $6, 7, 8, 9$ are all minimal generators: a sum of two positive elements is at least 12. The six consecutive integers from 12 through 17 belong to S, since

$$12 = 6 + 6, \quad 13 = 6 + 7, \quad 14 = 6 + 8, \quad 15 = 6 + 9, \quad 16 = 8 + 8, \quad 17 = 8 + 9.$$

Adding 6 repeatedly proves that every integer at least 12 belongs to S. Consequently its nonnegative gaps are exactly

$$\mathbb{N} \setminus S = \{1, 2, 3, 4, 5, 10, 11\}, \qquad F = 11. \tag{4}$$

For a gap f, testing $f + g_i \in S$ at the minimal generators suffices for the pseudo-Frobenius condition: every positive element of S is a generator plus an element of S. Both 10 and 11 pass this test, since their sums with generators are at least 16. Each smaller nonnegative gap fails:

$$1 + 9 = 10, \quad 2 + 8 = 10, \quad 3 + 7 = 10, \quad 4 + 6 = 10, \quad 5 + 6 = 11.$$

Negative integers contribute no pseudo-Frobenius numbers. If $f < -6$, then $f + 6 < 0$. If $-6 < f < 0$, then $f + 6 \in \{1, \ldots, 5\}$. Finally, $-6 + 7 = 1$. These are all gaps. This proves the asserted full set $\mathrm{PF}(S) = \{10, 11\}$.

In the generator order $(6, 7, 8, 9)$, take the NG-vector $(p_1, p_2, p_3, p_4) = (11, 10, 10, 10)$. All tests in (1) are displayed in Table 1; every entry is a generator of S.

g_i	p_i	$g_i + p_i - 10$	$g_i + p_i - 11$
6	11	7	6
7	10	7	6
8	10	8	7
9	10	9	8

Table 1: The complete nearly Gorenstein test.

Almost symmetry fails at $a = 1$: both 1 and $F - 1 = 10$ are gaps, and $1 \notin \mathrm{PF}(S)$.

3 Generation and minimality of the defining ideal

Put $R = \mathbb{Q}[x, y, z, w]$, and write $\varphi : R \to \mathbb{Q}[t]$ for the map in Theorem 1. Let J be the ideal on the right of (3). Each displayed binomial maps to zero, so $J \subseteq I_S$.

3.1 A terminating reduction proves generation

Use the four replacements

$$y^2 \longrightarrow xz, \qquad yz \longrightarrow xw, \qquad z^2 \longrightarrow yw, \qquad w^2 \longrightarrow x^3. \tag{5}$$

Give (x, y, z, w) the auxiliary weights $(1, 4, 6, 7)$. These are separate from their semigroup degrees $(6, 7, 8, 9)$. The four replacements strictly decrease auxiliary weight, respectively by $1, 2, 1, 11$. Each sequence of replacements therefore terminates.

A terminal monomial is divisible by none of y^2, yz, z^2, w^2. It has the form

$$x^a b, \qquad a \geq 0, \qquad b \in \{1, y, z, w, yw, zw\}. \tag{6}$$

The semigroup degrees of the six choices for b are

$$0, 7, 8, 9, 16, 17,$$

whose residues modulo 6 are all distinct. If two terminal monomials have the same image under φ, reduction of their degrees modulo 6 first identifies their choices of b and then identifies their exponents a. Thus all terminal monomials have distinct images, which are linearly independent powers of t.

For any $P \in I_S$, reduce its monomials using (5), collecting terms at the end, to obtain a linear combination N of terminal monomials. Every replacement changes P by an element of J, so $P - N \in J$ and $\varphi(N) = 0$. The independence just proved gives $N = 0$. Hence $P \in J$, proving $I_S = J$ and $\rho(S) \leq 4$.

3.2 Four relations are necessary

Use the positive grading $\deg(x, y, z, w) = (6, 7, 8, 9)$. The complete list of monomials of degree less than 14 is

$$1, x, y, z, w, x^2, xy,$$

with distinct degrees $0, 6, 7, 8, 9, 12, 13$. Therefore I_S has no nonzero homogeneous part below degree 14. For $\mathfrak{m} = (x, y, z, w)$, the ideal $\mathfrak{m}I_S$ has no nonzero homogeneous part below degree $20 = 6 + 14$.

The four binomials in (3) have distinct degrees $14, 15, 16, 18$, all below 20. Their classes in the graded $\mathbb{Q}$-vector space $I_S/\mathfrak{m}I_S$ are nonzero and linearly independent. Any ideal-generating set spans this quotient, since its polynomial coefficients reduce to constants modulo $\mathfrak{m}$. Every such set therefore has at least four elements. Together with the generation proof, this proves $\rho(S) = 4$ and completes Theorem 1.

4 An independent exact check by factorization graphs

For $n \in S$, let

$$Z(n) = \{(a, b, c, d) \in \mathbb{N}^4 : 6a + 7b + 8c + 9d = n\}.$$

Join two distinct factorizations by an edge when they share a positive coordinate, and denote the number of connected components by $c(n)$. The number of minimal relations of degree n is $c(n) - 1$; this is the standard presentation criterion recalled in [3, Theorem 3]. Here is the algebraic reason. The degree-n part of I_S consists of linear combinations of monomials in $Z(n)$ whose coefficients sum to zero. The degree-n part of $\mathfrak{m}I_S$ is spanned by differences of monomials sharing a variable, hence by differences within graph components. Its quotient therefore has dimension $c(n) - 1$.

Lemma 2. *For a numerical semigroup with four ordered minimal generators $g_1 < g_2 < g_3 < g_4$ and Frobenius number F, every disconnected factorization graph has degree at most $F + g_3 + g_4$.*

Proof. If $n > F + g_3 + g_4$, then $n - g_i - g_j > F$ for every distinct pair i, j. Thus there is a factorization of n containing both generators g_i, g_j. Given any two factorizations, choose a generator from each support. If the chosen generators coincide, the factorizations are adjacent; otherwise a factorization containing both connects them by a path of length at most two. The graph is connected. □

For the present example the bound is $11 + 8 + 9 = 28$. Exact enumeration of all factorization graphs through degree 28 finds precisely the four disconnected graphs in Table 2. All four have two components, so the independent count is $\rho(S) = 1 + 1 + 1 + 1 = 4$.

The Apéry set, consisting of the smallest elements in each residue class modulo 6, is $\{0, 7, 8, 9, 16, 17\}$. It agrees with the degrees of the six terminal basis monomials in (6) and with [3, Example 8].

Degree	First factorization	Second factorization	Components
14	y^2	xz	2
15	yz	xw	2
16	z^2	yw	2
18	w^2	x^3	2

Table 2: All disconnected factorization graphs; each listed fiber has exactly the two shown monomials.

5 Reproducibility and attribution

The accompanying package contains two standard-library Python programs. From the directory containing `src/` and `certificates/`, run

```
python src/proof_verify.py
python src/search_verify.py
```

The first checks the finite data of the terminating-reduction and minimality arguments. The second independently enumerates all factorizations through the proved bound, checks the semigroup hypotheses, and compares the complete result with the supplied JSON certificate. Both use exact integer arithmetic. The preceding proofs supply the universal steps beyond these finite checks.

This note and its programs were prepared by GPT-6 Astra. The result is a self-contained verification of a known example and its negative answer to the precise formulation (2). The source comparison and the literature search are documented in the accompanying literature review.

References

[1] A. Moscariello and A. Sammartano, *Open problems on relations of numerical semigroups*, arXiv:2406.00790v3, September 28, 2026, Problem 23. https://arxiv.org/abs/2406.00790v3.

[2] A. Moscariello and F. Strazzanti, *Nearly Gorenstein vs Almost Gorenstein Affine Monomial Curves*, Mediterranean Journal of Mathematics **18** (2021), article 127, Proposition 3.4 and Question 4.3. doi:10.1007/s00009-021-01761-1.

[3] T. Gomes, C. O'Neill, and E. Torres Davila, *Numerical Semigroups, Polyhedra, and Posets III: Minimal Presentations and Face Dimension*, Electronic Journal of Combinatorics **30**(2) (2023), #P2.57, Theorem 3 and Examples 8 and 18. doi:10.37236/10380.